

Phishing Between Loneliness and Skepticism: How Social Engineering Exploits the Need for Belonging, Safety, and Control

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Problem

Phishing remains the dominant entry point to serious cyber incidents, yet technical controls and awareness campaigns have not eliminated it. In Norway, public skepticism toward unknown senders has increased, while loneliness is elevated among young adults and students (SSB, 2025; NSO/Khrono, 2024). The tension is central: phishing often does not resemble a stranger attack. It imitates belonging, safety, order, status, help, and control.

Research problem: Why does phishing remain effective when people are both more lonely and more skeptical - and which emotional and contextual factors make individuals in higher education especially vulnerable?

Research question

Main RQ: How can phishing remain effective in a context of increased loneliness and increased general skepticism, and which emotional, cognitive, and contextual factors make individuals especially vulnerable to social engineering?

Hypotheses (summary): H1 loneliness correlates with susceptibility to belonging-framed lures; H2 skepticism reduces stranger-framed susceptibility; H3 situational pressure mediates; H4 verification habits and security culture are protective (PMT-informed).

Method (planned)

Element	Plan
Design	Exploratory sequential mixed methods
Sample	120-180 survey + 12-18 interviews; HE in Norway, 18+
Instruments	UCLA loneliness (short form), skepticism scale, scenarios S1-S6, semi-structured interviews
Analysis	Regression/mediation (quant); reflexive thematic analysis (qual); integrated joint display
Ethics	NSD/REK before collection; no deceptive live phishing; Appendices F-G

Contribution and relevance

- Human-centred model linking loneliness, skepticism, and lure type (beyond "weakest link").
- Empirical insight for Norwegian higher education security culture and OsloMet recommendations.

- Feasible MSc scope: behavioural study without classified infrastructure access.

Timeline (outline)

Ethics and instrument finalisation (Jan 2027) · Pilot (Feb) · Main survey and interviews (Mar-May 2027) · Analysis and writing (Jun-Nov 2027) · Submission (Dec 2027). Full timeline in thesis Appendix E.

One-page summary for supervisor alignment. Full draft: [master_thesis_draft.html](#) v0.6.